

- 7 (a) State the sum of the series  $1 + w + w^2 + w^3 + \dots + w^{n-1}$ , for  $w \neq 1$ . [1]

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- (b) Show that  $(1 + i \tan \theta)^k = \sec^k \theta (\cos k\theta + i \sin k\theta)$ , where  $\theta$  is not an integer multiple of  $\frac{1}{2}\pi$ . [2]

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- (c) By considering  $\sum_{k=0}^{n-1} (1 + i \tan \theta)^k$ , show that

$$\sum_{k=0}^{n-1} \sec^k \theta \sin k\theta = \cot \theta (1 - \sec^n \theta \cos n\theta),$$

provided  $\theta$  is not an integer multiple of  $\frac{1}{2}\pi$ . [5]

[illegible]

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- (d) Hence find  $\sum_{k=0}^{6m-1} 2^k \sin\left(\frac{1}{3}k\pi\right)$  in terms of  $m$ . [2]

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